



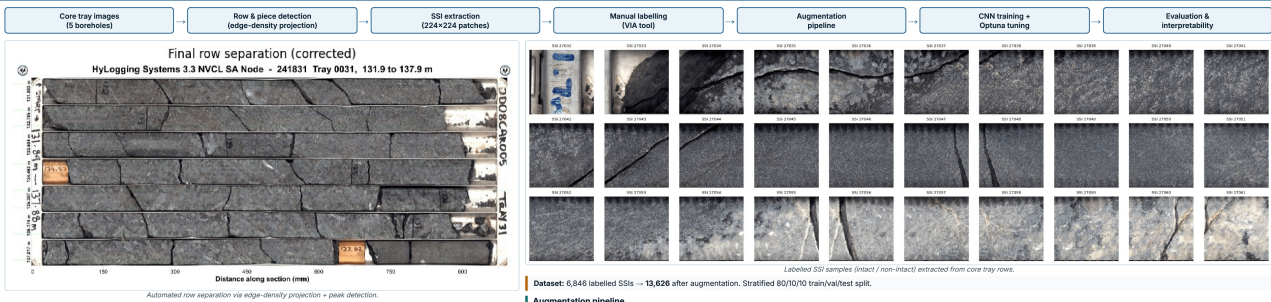
Automated Fracture Detection in Rock Cores Using Convolutional Neural Networks

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SURGE RESEARCH INTERNSHIP

Introduction

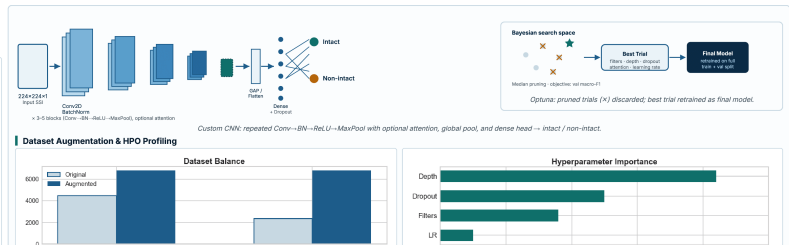
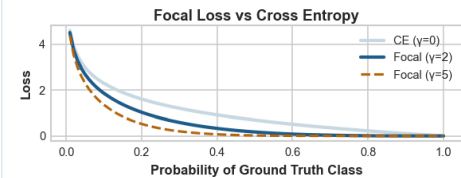
Rock Quality Designation (RQD) is a fundamental geotechnical index for rock mass classification in mining, tunnelling, petroleum, and foundation engineering, computed from the fraction of intact core pieces >10 cm per borehole run, and used in RMR and Q classification systems to govern excavation and support design. Manual RQD logging from core-tray photographs is **slow, labour-intensive, and subject to inter-logger variability**, especially for hairline or drilling-induced fractures. With drilling programs generating thousands of images per site, an automated, scalable, image-based approach is essential.

Methodology



Model & training

Custom CNN: Conv2D → BatchNorm → ReLU → MaxPool blocks on 224x224 grayscale SSIs with optional self-attention; He-normal init; L2 regularisation. Trained with **focal loss + label smoothing + per-class weighting**. Hyperparameters (filters, depth, dropout, attention) tuned via **Optuna** Bayesian search + median pruning, optimising validation macro-F1. Early stopping and LR-on-plateau used throughout.



Results & Analysis

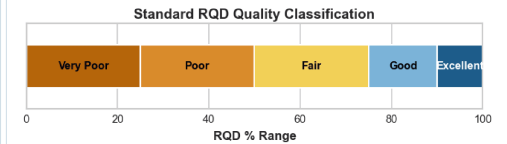


Applications in Engineering

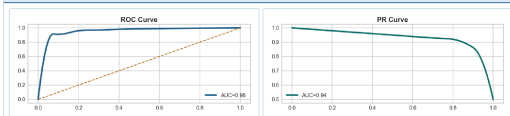
Automated fracture detection applies to **mineral exploration, petroleum reservoir characterisation, and tunnelling**. Automating intact vs. non-intact statistics extraction accelerates geological logging, freeing experts for complex lithological interpretation rather than manual hairline fracture counting.

Discussion & RQD Impact

The model reliably identifies intact core (recall 96-98%) but occasionally misses subtle fractures (recall 82-85%). However, **precision for non-intact is extremely high (91-96%)**. Because manual logging introduces subjective disagreement, this consistent bias is useful: the model acts conservatively, producing **safe-side RQD over-estimates** rather than dangerous under-estimates.



Classification Profiles



Conclusion

- Pipeline validated:** 94% val accuracy, 91% test accuracy on 685 held-out SSIs.
- Robust segmentation:** Sobel filter and horizontal projection separate core rows.
- Architecture:** GATMAP pooling captures global texture and localised edges.
- Deployment:** Conservative precision/recall tradeoff proves engineering readiness.

Future Work

- Optuna:** Complete full 50-trial search with focal-loss to push recall above 85%.
- Generalisation:** Test on remaining boreholes to validate cross-site adaptability.
- RQD Aggregation:** Implement core-run RQD calculation directly from SSI predictions.
- Transfer Learning:** Evaluate pretrained backbones with custom CNN.

Open Science & Reproducibility

All code for training the Custom CNN, Optuna HPO, dataset augmentation pipeline, and RQD estimation inference is completely open-sourced. The annotated rock core dataset (6,816 patches) is available for academic benchmarking.

References

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Acknowledgements

The author thanks the SURGE Program, IIT Kanpur for funding and research support, the project mentor and lab members for annotation guidance, and NVCL, Geoscience Australia for Hylogging data.